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Milestone Report - Group 20

1. Introduction

Stance detection is an important Natural Language Processing task that focuses on identifying whether a text expresses support, opposition, or remains neutral toward a specific target. For this project, the 'SemEval-2016 Stance Dataset' by Saif M. Mohammad, Parinaz Sobhani, and Svetlana Kiritchenko was chosen considering its relevancy and diverse target topics. It is a dataset consisting of tweets which are annotated based on their stance on the target topics, target of opinion and polarity, providing valuable data on the subject. In this project, the SemEval-2016 Stance Dataset will be utilized for developing accurate stance detection models. Then we will compare our results with the existing models that use the same dataset.

Up to this point in the project, the dataset was selected, the project roadmap and workload distributions were discussed and determined. We are ready to start with the preprocessing of the data.

2. Dataset Selection

The project uses the SemEval-2016 Stance Dataset which is designed for stance detection tasks. This dataset consists of labeled tweets with target topics consisting of climate change, atheism, Donald Trump, Feminist Movement, Hillary Clinton, and the legalization of abortion.
Each tweet is annotated with a stance, which are "favor", "against", or "neither", for a single target topic. Whether the target of opinion is the same as the given target topic is also included alongside the polarity/sentiment of the tweet. Hillary Clinton has the largest number of tweets with 984 tweets while climate change has the smallest with 564. All topics except climate change have more "against" tweets than the other stances while climate change has a majority of "favor" tweets.

Therefore, the dataset is a very good example for studying stance detection. The difference distributions of stances among topics will allow us to observe how the distribution of training data will impact the accuracy of our models. As SemEval-2016 Stance Dataset was used in the SemEval-2016 Task 6 competition, we will also have the opportunity to compare our results with already existing models.

Why This Dataset?

We will be using the SemEval-2016 Stance Dataset. It fits best with our objective and methodology for the following reasons:

- **Relevance:** It has clear benchmarks and full compatibility with the task objectives because of the particular design of this dataset for stance detection.
- **Diversity:** The dataset contains data from various domains; it hence allows for good generalization. It is also very important in training robust models.
- **Structure:** The dataset is well-structured and well-balanced like 50% Against, 25% Favor, and 25% Neutral. This dataset provides an almost balanced approach during training and evaluation, which reduces the possibility of overfitting during testing, especially for F1 scores.

Dataset Split

The dataset was divided into three parts which are the training set, development set, and test set. Fixed random seed was used in order to provide a fair and reproducible evaluation. SemEval-2016's original test set was preserved, while the training set was re-split into 60% for training and 10% for validation, with the test set remaining at 30%.

- **Training Set (60%):** This set will be used for model training and preprocessing.
- **Development Set (10%):** This set will be used to fine tune hyperparameters and select the best model.
- **Test Set (30%):** This set will be used for final evaluation.

This splitting methodology makes experiments reproducible while preserving the integrity of test data. It also allows for reliable metrics for performance comparisons as we can directly compare our results with the results of those who also used the SemEval-2016 dataset.

3. Approach Plan

As we are using the SemEval-2016's stance dataset, we will start by implementing the model with the highest F-score in SemEval-2016 Task 6 in an effort to reproduce its results. The highest F-score belonged to the organizers of the Task 6, who used a SVM with n-grams to reach a F-score of 68.98 (Mohammad et al., 2016). In a later publication, they improved their result slightly by using a linear-kernel SVM classifier that used word and character n-grams, and Word2Vec Skip-gram word-embedding features drawn from additional unlabeled data. They trained separate models for each of the categories for stance detection, then evaluated their models using the F-score, which is the harmonic average of F1 scores for the two main classes "favor" and "against". They compared the micro and macro F-scores, observing that micro F-scores were higher than the macro F-scores of all models, as their models performed better with target classes that were more frequent compared to classes that were less frequent. They compared the F-scores of the various SVMs they trained with different features and found that the SVM that used n-grams, the presence of the target of interest in the tweet, and the Word2Vec

Skip-gram word embeddings as its features performed best with the micro F-score of 70.3 (Mohammad et al., 2017), which is higher than the 68.98 they got in 2016 .

The highest F-score among participants of the SemEval-2016 Task 6 belonged to the team MITRE with F-score of 67.82 (Mohammad et al., 2016), who used transfer learning with two recurrent neural networks for each target class. The first recurrent neural network was used to predict related hashtags while the second recurrent neural network was used for stance classification using the predictions of the first neural network as additional features (Zarella & Marsh, 2016). As the score of the SVM is higher than the recurrent neural network's despite its simpler architecture, we will attempt to only replicate the work of Mohammed et al. Then we will try to improve their result by using methods that were developed after 2017 like BERT. We will use BERT with SVM like Mohammad et al., but also with a Naive Bayes model, a Feedforward Neural Network and a Recurrent Neural Network. We hope that using newer word embedding methods like BERT will lead to higher F-scores and we also hope to see if a particular model will outperform the other models that also use the BERT word embeddings.

4. Next Steps

Further steps involved in the project will be the development of the baseline SVM model which includes word and character n-grams, and Word2Vec Skip-gram word embeddings to set a performance benchmark. Once our group successfully reproduces the model mentioned above, the next step will involve the implementation of BERT-based models to enhance the results obtained with the original SVM.

Workload Division:

Our group has organized the workload to ensure smooth progress and clear responsibilities for everyone:

- Report writing, presentation preparation and data preprocessing will be shared among group members to keep the workload balanced.
- Bengi will develop the SVM model that will reproduce the results of Mohammad et al. and establish a performance baseline, then will use the BERT word embeddings instead of Word2Vec embeddings with the SVM to hopefully get a better F-score.
- Serhan will use the BERT word embeddings with recurrent neural networks.
- Nida will use the BERT word embeddings with Naive Bayes.
- Zeynep will use the BERT word embeddings with feedforward neural networks.

Multiple models that satisfies project objectives will be produced by following the procedures described in this method and integrating individual contributions and teamwork. Then we will compare the F-scores of our models alongside themselves and the models in the literature.

5. REFERENCES

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